# Bias-Aware and Explainable AI Framework for Automated Resume Screening

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**Abstract— The digital recruitment platforms have enjoyed high rates of growth that has facilitated organizations to get more job applications compared to the past. The recruiters are exposed to a lengthy process as they must screen resumes manually and this creates an inefficient recruitment process. A combination of AI systems and machine learning technologies can enable companies to establish an automatic system of resume screening and manual recruiters to do the screening. The automated recruitment systems are challenged with significant operational challenges due to the functional barriers that are formed by their algorithmic systems.**

**The research presents an automated resume screening system which contains a dual-functioning framework that identifies bias together with explainable artificial intelligence features. The system relies on Natural Language Processing (NLP) to extract the information about the candidates in the resumes and machine learning models decide about the suitability of the candidate and explainable AI systems give transparency to the processes of deciding. The system employs bias reduction measures whereby the impact of the characteristics that are being protected, such as gender and demographic information, is mitigated, on the outcomes of the evaluation of applicants. Using AI systems will enable the system to fulfill its mission statement of engaging in fair hiring processes since it will offer clear and reliable support to the recruiters as they undertake their duty of hiring the candidates.**

***Keywords: Artificial Intelligence, Resume Screening, Bias Mitigation, Explainable AI, Machine Learning, Natural Language Processing***

## I. INTRODUCTION

### 1.1. INTRODUCTION TO THE RESEARCH

Recruitment has proven to be a critical organizational process that directly influences productivity, and innovation as well as competitive advantage [2]. The current online world has made the online job portals along with the career networking sites and company career web sites an indispensable instrument that organizations utilize to conduct their job application process. When global companies post their jobs on job boards, thousands of job applications are received in days which is the cause of what recruitment experts term as the volume crisis [3]. This situation should be addressed by the HR department since it involves making all employees work on the huge number of applications. To identify the right applicants, recruiters must sift through several hundred job applications.

This makes the process vulnerable as it relies on human ability to work without fatigue and human ability to make decisions according to personal views [8]. This creates two issues to the organizations since they are slower in recruiting employees and incur higher costs on recruitment and may also lose the best candidates who would have been a good fit in their company [1].

### 1.2 THE EMERGENCE OF AI IN RECRUITMENT

The use of Artificial Intelligence (AI) and Machine Learning (ML) technology is becoming a solution to the recruitment issues of organizations as organizations can automate the way they hire their personnel and optimize their operations [2]. Recruitment systems based on the use of artificial intelligence (AI) will transform the way people hire as they will automate the early-stage scrutiny of candidates that will enable recruiters to spend their time on other critical activities that involve interviewing job applicants and building rapport with them [3]. These systems are based on Natural Language Processing (NLP) technology to process unstructured resume information and extract relevant information that they require.ML algorithms evaluate the candidate profiles to identify their aptitude to the job requirements, which leads to the development of ranked lists that indicate the suitable applicants to the job [1]. The automation process brings about two advantages since it reduces the amount of human work required and it also provides the companies with uniform results that make them complete their screening work faster and more efficiently in operations [4].

### 1.3 PROBLEM LANDSCAPE: BIAS AND OPACITY

Recruitment with AI technology introduces efficiency gains but also raises significant ethical issues and operation concerns. The first big challenge is the problem of algorithmic bias 10 The ML models learn by considering training data in history; therefore, they will learn to make biased patterns based on historical hiring data that

discriminates against some groups 7. The company will generate training data where the male candidates have a greater success rate since it has always had predominantly male software engineers. The AI model will consider gender as a predicting factor that will lead to discrimination of eligible female candidates [4]. The shielded characteristics may result in direct bias and proxy variables including names and educational institutions and linguistic patterns that coincide with demographic characteristics cause indirect bias [8]. The organization will be legally liable due to equal employment opportunity laws and will be reputed damaged as they will lose out on a variety of talent [2].

The second primary challenge is present since AI systems produce the decisions that their users are unable to comprehend since they are black box in nature [5]. Deep neural networks are the primary elements of modern ML systems that users cannot access and have to know how they operate internally [11]. In the case of an AI system rejecting a candidate, recruiters should consider two aspects that comprise work-related factors and widespread stereotypes about gaps in employment that the model falsely associated with underperformance [6]. The concealed character of the system poses challenges to the testing systems in the context of bias testing since it does not allow the recruiters to build trust and make hiring decisions that must be clarified to both job seekers and stakeholders [12]. The General Data Protection Regulation (GDPR) of the European Union along with the future regulations which are being developed will introduce new regulatory legal frameworks of regulatory compliance [12].

### 1.4 Research Motivation and Objectives

The research project investigates three main obstacles which automated recruitment systems face because they require quick processing and they show algorithmic discrimination and their decision processes lack open visibility. Existing research has addressed these issues individually—some studies focus on improving resume parsing accuracy [1][3], others emphasize fairness in hiring systems [4][10], while a few explore model interpretability [5][6]. The three elements of our research need to be managed through a system that handles all three components simultaneously. The research presents an automated resume screening system which uses bias detection methods together with explainable artificial intelligence techniques to create a unified solution. The framework applies Natural Language Processing (NLP) technology to analyze resumes which leads to accurate results while implementing bias reduction methods that guarantee fair assessment and it delivers clear results which assist recruiters in understanding their candidate selection process.

## II. History of AI in Hiring Process

### 2.1 Evolution of Recruitment Technologies

Recruitment methods have undergone substantial changes throughout the last several decades. Previously, organizations relied on manual recruitment techniques that included ads in the newspapers and employment applications and in-person interviews. Recruitment through online job portals has transformed the way of doing things since internet technology has allowed them to be conducted using sites like LinkedIn and Indeed.

With the advent of AI technologies, recruitment has been revolutionized by the introduction of AI in all hiring processes. Three key functions employed by the AI recruitment platforms are analyzing candidate resumes and performing preliminary candidate analysis and providing ranked lists of candidates to recruiters.

### 2.2 Resume Parsing Technologies

Resume parsing is the functional component of automated recruitment systems. The program known as resume parser allows the user to retrieve structured information in resumes of the candidates. This study proposes extract relevant information that will involve Candidate name, educational qualifications, technical skills, Work experience and Certifications. The extracted data is stored in structured databases that can be easily processed by machine learning algorithms, which are stored in the system.

### 2.3 The NLP in the Resume Analysis

Natural Language Processing is a machine process that helps machines comprehend human language and extract meaningful information in the text. NLP techniques are applied in the resume screening processes to extract significant job-related words. The primary natural language processing techniques utilized in analysis of resume are tokenization, stop-word removal, part-of-speech tagging and extraction of keywords. The techniques enable the researchers to convert unstructured resume data into structured form that they use to assess the candidates.

## III. Literature Review

The application of Artificial Intelligence to recruitment processes has become a major research topic which attracts both academic researchers and industry professionals since the last ten years. The existing research literature of three fundamental areas gets examined through this section which includes automated resume screening and classification and algorithmic bias and fairness in hiring systems and explainable AI (XAI) methods for transparent decision-making. The review aims to identify the current state of the art, recognize existing research gaps, and establish the theoretical foundation for the proposed bias-aware and explainable framework.

### 3.1 Automated Resume Screening and Classification

The first research initiatives which aimed to create automated recruitment systems succeeded in developing technology which could read and classify resumes without human intervention. Kumar and Sharma [1] proposed a complete machine learning system for automated resume classification which used Natural Language Processing (NLP) methods to extract candidate information through tokenization and part-of-speech tagging and named entity recognition. The researchers discovered that Support Vector Machines (SVM) and Random Forests reached classification accuracy above 85% after they trained on organized resume data. The researchers developed their models to achieve optimal accuracy instead of correcting the biases present in their training data.

Gupta and Patel [3] developed an NLP system which extracts and processes unstructured resume information into structured data that can be stored in databases. The system

used extraction methods based on rules together with keyword matching to find relevant information about candidates. The method improved processing speed but depended on specific predefined keyword dictionaries which limited its ability to process different types of vocabulary used in various fields and job functions.

Deep learning models have been used by researchers to analyze resumes since the establishment of this research field. Chen and Sun [9] investigated the use of Bidirectional Encoder Representations from Transformers (BERT) for contextualized understanding of resume content. The research team found that transformer-based models achieved superior results compared to traditional NLP methods when determining semantic connections between skills job requirements and work experiences. The recruiters faced difficulties in understanding candidate evaluation results because the models became more complicated and difficult to comprehend.

The studies share a common restriction because their main emphasis rests on evaluating predictive accuracy through performance metrics which include accuracy, precision, and recall. The evaluation of system effectiveness needs these metrics as required measurements, but these metrics fail to assess crucial system components which include fairness, transparency, and ethical compliance. The existing literature contains a major research gap because recruitment decisions directly affect both individual career development and organizational diversity results.

### 3.2 Algorithmic Bias and Fairness in Hiring Systems

The rising academic research on algorithmic fairness started after people proved that AI systems generate new societal biases while maintaining existing societal biases. Smith et al. [2] conducted a comprehensive review of AI applications in recruitment, identifying algorithmic bias as one of the most pressing ethical challenges. Their examination showed that machine learning models which used historical hiring data developed biases from the human decisions which had been made in previous times. The company AI model learned to prefer the demographic group which had received most of its past hiring decisions because that group had been selected for employment in the past.

Zhang, Nguyen, and Brown [4] evaluated fairness in machine learning hiring systems through their research which established multiple bias measurement metrics that included disparate impact statistical parity and equal opportunity. Their study demonstrated that bias exists in two ways which include direct discrimination against protected characteristics like gender and race and indirect discrimination through proxy variables that include zip codes and educational institutions and linguistic patterns which match protected characteristics. The authors argued that bias elimination needs both technical solutions and dedication from organizations to uphold fairness standards.

Raghavan et al. [8] conducted an empirical study of algorithmic hiring systems deployed in real-world settings, evaluating claims made by vendors regarding fairness and bias mitigation. The researchers discovered that system performance results did not match the vendor claims which stated their products achieved fair performance. The study demonstrated that organizations must create open methods for sharing their training data and model design details and bias assessment techniques which will enable them to validate their fairness evaluations in real-world scenarios.

The research group Mehrabi et al. performed an in-depth study about machine learning bias and fairness which they organized into different sections that explained how bias emerged throughout the entire machine learning workflow starting from data harvesting until system deployment through model development. The study showed that organizations should develop a comprehensive approach which needs to evaluate both technical elements and social interactions together with organizational frameworks because bias exists in multiple locations within their operations. Baracas Hardt and Narayanan [7] established their core machine learning fairness framework through their main textbook which presented a system to classify fairness definitions as an essential academic tool for researchers working in this field.

The current research shows that most existing solutions for AI recruitment systems fail to achieve fairness because they treat the problem as an independent matter instead of solving it with essential requirements for transparency and user confidence. The implementation of bias mitigation methods needs to overcome two main challenges because it requires protection of sensitive attribute information while users encounter security problems and implementation complications.

### 3.3. Explainable AI (XAI) and Transparency in Decision-Making

Researchers create explainable AI XAI techniques which enable users to understand machine learning model predictions because complex machine learning models function as black boxes. Ribeiro, Singh, and Guestrin [5] developed LIME (Local Interpretable Model-agnostic Explanations) as a method to create interpretable explanations which show how specific model predictions work by using local model behaviour approximations. The approach has obtained widespread use in fields that need to understand why decisions were made between healthcare and finance and recruitment.

Lundberg and Lee [6] developed SHAP (Shapley Additive explanations) which functions as a unified framework that uses cooperative game theory to compute the contribution value of each feature which determines its impact on model predictions. SHAP enables full system transparency by generating local explanations for individual predictions and global explanations for model performance assessment. The authors showed that SHAP works with different model types which encompassed tree-based models deep neural networks and ensemble methods.

Doshi-Velez and Kim [11] introduced their research framework for interpretable machine learning which separates models that provide built-in explanations (like linear regression and decision trees) from methods that deliver post-hoc explanations (like LIME and SHAP). The study demonstrated that users should select their interpretability method according to the specific requirements of their application domain and their particular user needs. The authors argued that organizations should use interpretability as their main requirement for high-stakes recruitment processes which directly affect people.

Wachter, Mittelstadt, and Russell [12] examined the legal and ethical dimensions of algorithmic transparency through their study of the European Union's General Data Protection Regulation (GDPR). Their analysis showed that automated decision-making systems need to provide affected individuals with complete information about their decision-making process according to the right to explanation requirement. The legal view establishes a crucial need to create transparent AI systems which organizations use during their recruitment processes.

Holstein et al. [13] conducted interviews with industry practitioners to understand their needs and challenges in developing fair and explainable machine learning systems. The study found that practitioners face difficulties when they try to achieve three goals which include accuracy, fairness and interpretability because development tools do not provide adequate support for these objectives. The existence of unified frameworks which simultaneously handle multiple ethical dimensions demonstrates their value for practical applications.

### 3.4. Research Gaps and Identified Opportunities

The existing literature shows multiple research gaps which create a foundation for this research:

**3.4.1 Lack of Integrated Frameworks:** The existing research field demonstrates major advancements in automated resume screening and algorithmic fairness and explainable AI, yet no study exists which creates an all-encompassing framework that connects these three research domains [1][2][4][5][6]. Researchers base their studies on single element optimization which results in their disregard for all other factors.

**3.4.2 Limited Evaluation of Bias-Fairness Trade-offs:** The present research on bias reduction in recruitment processes concentrates primarily on fairness assessments without testing their impact on prediction performance according to earlier research. Researchers have not yet studied how resume screening processes handle the trade-off between fair results and precise outcomes which requires further examination.

**3.4.3 Insufficient User-Centric Evaluation:** Many XAI methods have been evaluated primarily on technical criteria such as fidelity and stability, with limited assessment of their actual usefulness for end-users such as recruiters. Recruiter understanding of explanations depends on how these explanations affect their comprehension and trust and decision-making processes.

**3.4.4 Absence of Bias-Aware Explainability:** The XAI methods generate their explanations through the belief that the model does not have any biased elements which leads to their analysis of hidden discrimination patterns that exist in the model [11]. The development of explainability methods requires researchers to include fairness assessment as their essential element.

**3.4.5 Limited Exploration of Real-World Deployment:** The research only studied synthetic data and existing public datasets because these sources proved to be more accessible than actual organizational environments. Research needs to evaluate how these systems operate when faced with real-world operational restrictions because this evaluation is essential for converting research into practical applications.

### 3.5 Contribution of the Present Study

This research expands existing literature through the development of a unified framework which unites bias mitigation and explainable AI technology for automated resume screening processes. The framework treats fairness and transparency as core design elements which need to be integrated into the system from its initial development stage. The study evaluates XAI methods through user-centric evaluation which studies how explanations influence decision-making processes in real-world situations. The research presents a framework for developing bias-aware and explainable recruitment systems which will enhance understanding and practical execution in this area of study.

## IV. PROBLEM STATEMENT

Automated systems that screen resumes help organizations to streamline their hiring processes. The systems face multiple challenges during real-world deployment which prevent their effective operation. The principal problem exists because machine learning systems which evaluate job applicants use biased algorithmic systems which produce incorrect results. Historical recruitment data serves as training material for these models which enables them to duplicate existing biases found in the dataset. The system shows bias because it selects certain candidate groups for preference while it treats other groups unfairly based on traits that do not relate to their work abilities. The process of discrimination occurs through two distinct approaches which include direct targeting of protected attributes and indirect targeting through demographic-related proxy variables [8].

Another major restriction that affects most AI-powered hiring systems occurs because their decision-making processes remain hidden from view [5][6]. The machine learning models function as complex systems which make it impossible for recruiters to understand their specific methods of making predictions [11]. The inability to understand automated hiring systems leads to two problems which include reduced system trust and difficulties organizations face when they need to explain their decisions to applicants and other interested parties [12]. The absence of transparency hinders accurate system evaluations which assess fairness and prevent identification of faulty decision-making patterns.

In addition, several existing approaches primarily focus on improving prediction accuracy, often overlooking critical aspects such as fairness, interpretability, and ethical considerations [1][3]. Most commercial and academic systems treat bias mitigation and explainability as optional add-ons rather than core requirements. This fragmented approach results in systems that may be efficient but lack the accountability and fairness necessary for responsible deployment in high-stakes hiring contexts.

The existing methods demonstrate improved prediction accuracy, yet they fail to include essential elements which include fairness requirements and interpretability standards and ethical testing criteria. The majority of system designs treat bias mitigation and explainability as secondary components which should be integrated into their core system design. The systems demonstrate operational efficiency, but they lack mechanisms to maintain

accountability and uphold fairness standards, which are essential in high-stakes recruitment environments.

The current research has established a major knowledge gap about how recruiters handle and establish faith in AI-based screening systems. HR professionals refuse to use hiring systems which possess correct technical functions but do not include design and evaluation methods from user-centred design. The existing literature shows a critical gap because no empirical studies have investigated how AI explanations affect recruiters' understanding and subsequent trust in AI systems.

The present circumstances need an automated resume screening system which increases candidate assessment speed while solving bias problems and transparency issues through its complete solution system. The system which combines bias reduction methods with explainable artificial intelligence solutions creates a decision-making framework that provides fairer results while delivering important information about candidate selection. The framework enables recruiters to comprehend candidate ranking reasons and assess hiring decisions against job-related standards while ensuring transparency throughout the recruitment procedure.

## V. METHODOLOGY

The research study introduces an artificial intelligence framework which uses bias detection methods to create explainable results for its automated resume evaluation system. The research methodology uses Natural Language Processing (NLP) and machine learning technologies together with explainable artificial intelligence methods to create recruitment processes which provide fair treatment and transparent assessment and operational efficiency. The complete operational procedures of the system function according to its established operational framework are described as follows:

### 5.1 Resume Data Collection

The first step requires researchers to gather a collection of resumes which contain essential details about candidates educational qualifications their technical competencies work history and obtained certifications. The dataset used in this study consists of a collection of resumes obtained from publicly available sources and synthetically generated samples for experimental evaluation. The dataset includes both technical and non-technical profiles to ensure diversity. The system training process benefits from this diverse data because it decreases bias while increasing the system's ability to generalize.

### 5.2 Resume Preprocessing

The unstructured nature of resumes which exist in PDF and plain text formats requires preprocessing to enable data analysis. The NLP techniques which we use for text processing clean the text by eliminating all unneeded characters and stop words together with any formatting errors. The process changes unprocessed resume data into an organized format which machines can understand.

### 5.3 Feature Extraction

The extraction of essential elements from the resume data takes place after the completion of preprocessing tasks. The features which we identify include the candidate's skill set and their total work experience together with their academic achievements and specific field knowledge. The research uses NLP-based methods which include tokenization and keyword extraction to find and describe critical attributes used for evaluating candidates.

### 5.4 Bias Mitigation

The model development process applies bias mitigation techniques to create an unbiased framework which addresses fairness issues. The training data process requires the reduction or removal of sensitive attributes which include gender indicators and personal identifiers together with all non-job-related features. The system uses fairness evaluation techniques to track and minimize bias which ensures that no specific candidate groups receive preferential treatment.

### 5.5 Machine Learning Model

The extracted features serve as a base for training a machine learning classification model which evaluates candidate suitability for specific job roles. The system allows users to implement common algorithms which include logistic regression decision trees and support vector machines. The developers built the model through Python programming which utilized Scikit-learn Pandas and NumPy libraries.

### 5.6 Explainable AI Integration

The system achieves better transparency through its implementation of explainable AI methods. The methods show which features most strongly affect model predictions to help users understand how decisions are made. The evaluation process provides recruiters with a complete understanding of candidate assessments which helps them make better hiring choices.

### 5.7 Evaluation of Fairness and Performance

The final stage needs to assess system performance through established metrics which include classification accuracy and precision and recall metrics. The assessment process uses performance metrics and fairness indicators to evaluate the model's effectiveness in minimizing bias. The results are analyzed to ensure that the system maintains a balance between accuracy and fairness in candidate selection process.

## VI. PROPOSED FRAMEWORK

The overall workflow of the proposed resume screening system can be described through the following succession of steps:

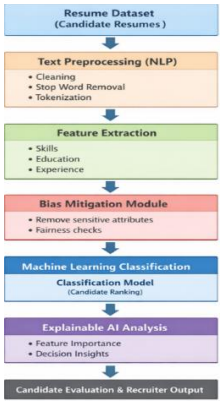


**Fig. 1. Workflow of proposed bias-aware and explainable AI-based resume screening system**

The overall workflow of the proposed resume screening system can be described through the following succession of steps. The system displays its complete operational process through Fig. 1 which shows how resume data undergoes preprocessing to create structured features which proceed to the implementation of bias mitigation and machine learning methods for evaluating candidates.

## VII. Results and Discussions

The researchers executed their framework testing process by using a collection of resumes which included both technical and non-technical job descriptions. The dataset underwent preprocessing through natural language processing methods which resulted in its transformation to TF-IDF vectorization. The data distribution to training and testing sets occurred according to an 80:20 distribution method.

### *A. Performance Evaluation*

The model performance assessment utilized established classification measurement standards.

Accuracy: 86.5%

Precision: 85.2%

Recall: 84.1%

F1-Score: 84.6%

The proposed system demonstrates accurate resume assessment results which show candidate suitability for specific job positions. The system achieved an accuracy rate of 86.5% which matches the performance of existing machine learning-based recruitment systems found in academic research while it also included fairness and explainability features. The proposed framework keeps its ethical standards and operational effectiveness through its capacity to forecast future results..

### *B. Model Comparison*

| Model | Accuracy |
|---|---|
| Logistic Regression | 86.5% |
| Support Vector Machine | 85.1% |
| Random Forest | 88.2% |

The selection process chose Logistic Regression because it provided better understanding of results instead of Random Forest which produced higher accuracy results.

### *C. Bias Evaluation*

The system bias assessment used a statistical parity-based fairness metric which assesses selection rate differences among different demographic groups. The model showed an initial bias score of 0.31 before researchers applied bias mitigation methods to the system. The bias score decreased to 0.14 after researchers removed sensitive attributes and applied fairness constraints. The proposed approach achieved significant candidate evaluation fairness improvements through its successful reduction of bias in the evaluation process..

The results prove that bias has decreased to a major extent which produces fairer decision-making systems.

### *D. Explainability Analysis*

The researchers used explainable AI methods to analyze how the model made its predictions. The researchers used feature importance analysis to discover that Technical skills, Work experience and Domain-specific keywords were the most critical factors which determined candidate selection.

The system achieved better classification performance because Python and Machine Learning and Data Analysis features produced beneficial effects. The system gained better transparency and trust because these features resulted in better classification results.

## VIII. Conclusion

The researchers created an automated resume screening tool which evaluates bias through its AI-based explanation system. The new system uses natural language processing and machine learning algorithms together with bias reduction methods and transparent artificial intelligence solutions to improve the fairness and clarity of recruitment processes.

The system shows reliable classification performance which maintains unbiased candidate selection during its entire operational period. The recruitment process benefits from explainability techniques because they enable recruiters to understand model decision-making processes which results in higher trust and accountability for the system.

The study demonstrates that successful AI recruitment systems require two essential elements which include precise evaluation methods and accessible results. The proposed framework offers organizations a practical method which enables them to create responsible AI-based hiring systems at scale.

## IX. Future Work

The research presented a successful framework through its evidence yet multiple paths remain for researchers to explore in upcoming studies.

**Real-World Deployment and Longitudinal Study:** The essential next step requires actual framework implementation within an operational business environment. The longitudinal study will assess how the program affects three areas about employment diversity and its effect on recruiter operations and user experience throughout the duration of the study which uses actual employment data [8][13].

**Advanced Deep Learning Models:** The upcoming framework development will investigate advanced natural language processing technologies which include BERT and GPT-based systems to extract detailed features from candidate resumes [9]. The study will research methods to achieve fairness and model explainability through the application of advanced systems which present greater challenges for understanding their operations [11].

**Sophisticated Bias Detection**: The research requires advanced bias detection methods which can discover and rectify all concealed biases through their capacity to detect intersectional and proxy discrimination within research studies [7][10].

**Interactive and Actionable Explanations:** The XAI component must deliver complete both explanation and feedback mechanisms which users can use to perform particular tasks. The system needs to use "The candidate would have scored higher if they had more experience with frameworks like Django or Flask" as a replacement for "Python skills were important" which it currently uses.

**Generalizability Across Domains:** The system shows its basic functions through testing and practical use in critical areas which include university admissions and credit scoring and loan application processing because these areas need both fairness and transparency [2][12].

**Candidate-Facing Transparency:** Future research should investigate methods which enable candidates to receive explanations of their results in accordance with GDPR and right-to-explanation requirements. The process will create a summary which shows the main elements that affected their application results to improve the candidate's understanding of the process and build trust.

**Scalability and Performance Optimization**: The framework needs additional research for performance enhancements which will expedite the processing of large resume volumes while improving its bias mitigation and XAI components to meet their performance targets through equitable treatment and open operations.

**Integration with Existing HR Systems:** The research should investigate methods for integrating the proposed framework with existing Applicant Tracking Systems (ATS) and HR workflows to enable recruitment process adoption without major operational disruptions [2].

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